

COMP4009 - Foundations of Computer Systems

Week 1 – Numbering Systems

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Week 1 & 2

- How is information stored in a computer?
- Binary arithmetic basics
- Reading these 2 weeks:
 - **Dale and Lewis Chapters 1-3**
 - **Forouzan and Mosharraf Chapter 1-3**
 - **Petzold, Code: The Hidden Language of Computer hardware and Software**
 - **First ten chapters**

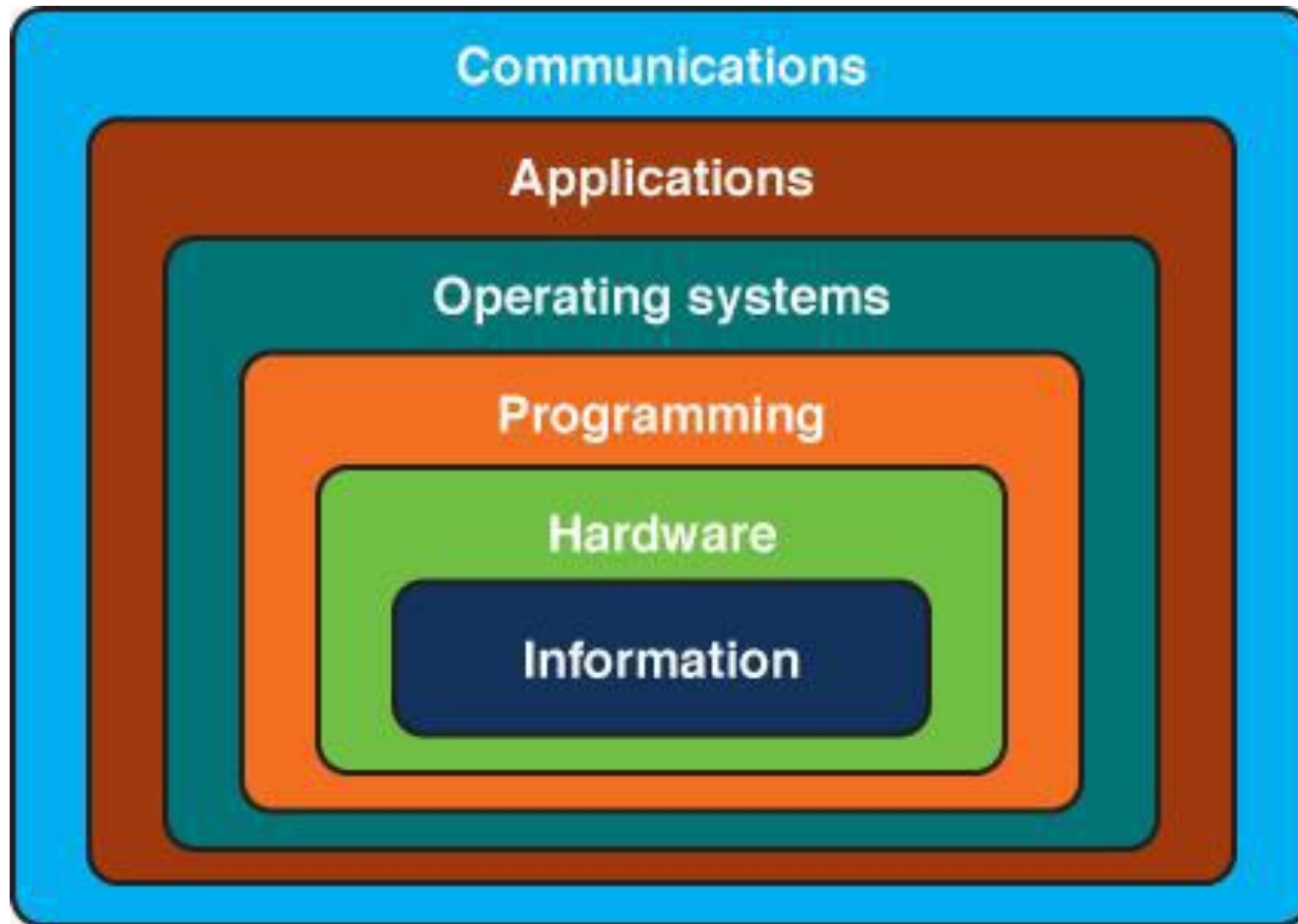
Learning Outcomes for Today

- General introduction to module organisation
- Introduction to binary arithmetic
 - Converting from one number system to another
- It is essential to understand binary numbers for this course, we will keep returning to it

Computing Systems

- **Hardware** The physical elements of a computing system (printer, circuit boards, wires, keyboard...)
- **Software** The programs that provide the instructions for a computer to execute

Layer of a Computing System



Layer of a Computing System

- **Information:** How the information is represented on a computer.
 - Information stored as binary numbers
- **Hardware:** Physical hardware of a computer system.
 - Components consist of gates and circuits that control the flow of electricity
- **Programming:** Software or instruction to manage and process data.
 - Low and High Level Programming language
- **Operating Systems:** Every computer has an operating system to help manage the computer's resources.
 - Windows, Linux, MacOS
- **Application:** Software developed to solve specific problems
 - Simulations, Information Systems
- **Communications:** Computer are connected to networks to share information and communicate with other computers
 - Networks, Internet

Information Layer

- Computers store information by using memory and on or off type of switches
- To understand how the computer stored information and manage this information we must learn binary notation
- Weeks 1-3 will be dedicated to the information layer of our computer

Binary

- Count numbers in binary
- Conversion from binary to another number base system (hexadecimal, decimal, octal)
- Basic arithmetic operations in binary
- It is tricky, so you should practise

Why Binary?

- Computers used electronic devices to store and manipulate information
- The electronic devices are like switches or bulbs that are either ON or OFF
- Computer only understands on and off information and this is translated into two numbers (or binary digits)

1 ON



0 OFF



Counting in Binary

- 0 -> 0
- 1 -> 1
- 2 -> 10
- 3 -> 11
- 4 -> 100
- 5 -> 101
- ...

Decimal	Binary
0	00000
1	00001
2	00010
3	00011
4	00100
5	00101
6	00110
7	00111
8	01000
9	01001
10	01010
11	01011
12	01100
13	01101
14	01110
15	01111
16	10000
17	10001
18	10010

Number Formats - Bases

- We use decimal numbers (base 10)
 - 10 different digits (0,1,2,3,4,5,6,7,8,9)

3241.56 base10

$$3 \times 10^3 + 2 \times 10^2 + 4 \times 10^1 + 1 \times 10^0 + 5 \times 10^{-1} + 6 \times 10^{-2} = 3241.56$$

- A number in another base (base 5)
 - 5 different digits (0, 1, 2, 3, 4)

3241.32 base5

$$3 \times 5^3 + 2 \times 5^2 + 4 \times 5^1 + 1 \times 5^0 + 3 \times 5^{-1} + 2 \times 5^{-2} = 446.68 \text{ base10}$$

Binary Numbers

- We have only two symbols or digits to represent binary numbers

0 1

- A binary number is a list of 1's and 0's
- To translate a binary number into the corresponding decimal form we have to multiply each digit for each corresponding power of 2

1011_{base 2}

$$\begin{array}{ccccccc} 1 \times 2^3 & + & 0 \times 2^2 & + & 1 \times 2^1 & + & 1 \times 2^0 = 11_{\text{base10}} \\ 8 & + & 0 & + & 2 & + & 1 = 11 \end{array}$$

Binary Numbers Stored in a Computer

- A binary digit is called bit
Bit = On or OFF = 1 or 0
- A group of 8 bits is a byte
- What is the biggest number that we can store in a byte?

11111111

$$1 \times 2^7 + 1 \times 2^6 + 1 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$128+64+32+16+8+4+2+1=255$$

Binary and Hexadecimal

- 4 bits can be used to represent the 16 different hexadecimal symbols
- Hexadecimal numbers are numbers in base 16
0 1 2 3 4 5 6 7 8 9 A B C D E F
- A byte is then equivalent to 2 hexadecimal numbers, or nybbles (1 nybble=4 bits).

3F4_{base16}

03F4 hex

= 0000 0011 1111 0100 binary

bin	dec	hex
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

Binary and Octal

- 3 bits can be used to represent the 8 different Octal symbols
- Octal numbers are numbers in base 8
0 1 2 3 4 5 6 7
- e.g. 3647_{base8}

3647_{Octal}

= 011 110 100 111 binary

Binary	Decimal	Octal
000	0	0
001	1	1
010	2	2
011	3	3
100	4	4
101	5	5
110	6	6
111	7	7

Why Do We Use These Number Systems?

- Binary
 - Same format as the computer hardware
 - Zero / One
 - On / Off
 - Can see bit patterns
 - 01100110...
- Hexadecimal
 - Easy to convert to binary
 - One hex digit = 4 bits
 - E.g. F4 (hex) =

1111 0100 (binary)

The diagram shows two arrows pointing from the binary groups to their hexadecimal equivalents. The first arrow points from '1111' to 'F'. The second arrow points from '0100' to '4'.

F 4

Conversion from Decimal to Binary

- Repeat division by 2 until the quotient is 0 and keep the remainder. Pass the quotient to the next stage, until it is zero
- E.g. 10 to binary?
- $10/2 = 5 \text{ r}0$
- $5/2 = 2 \text{ r}1$
- $2/2 = 1 \text{ r}0$
- $1/2 = 0 \text{ r}1$
- $10 \rightarrow 1010$
- So $10 = 8 \times 1 + 4 \times 0 + 2 \times 1 + 0 \times 1$ or 1010 binary

Conversion from Decimal to Binary

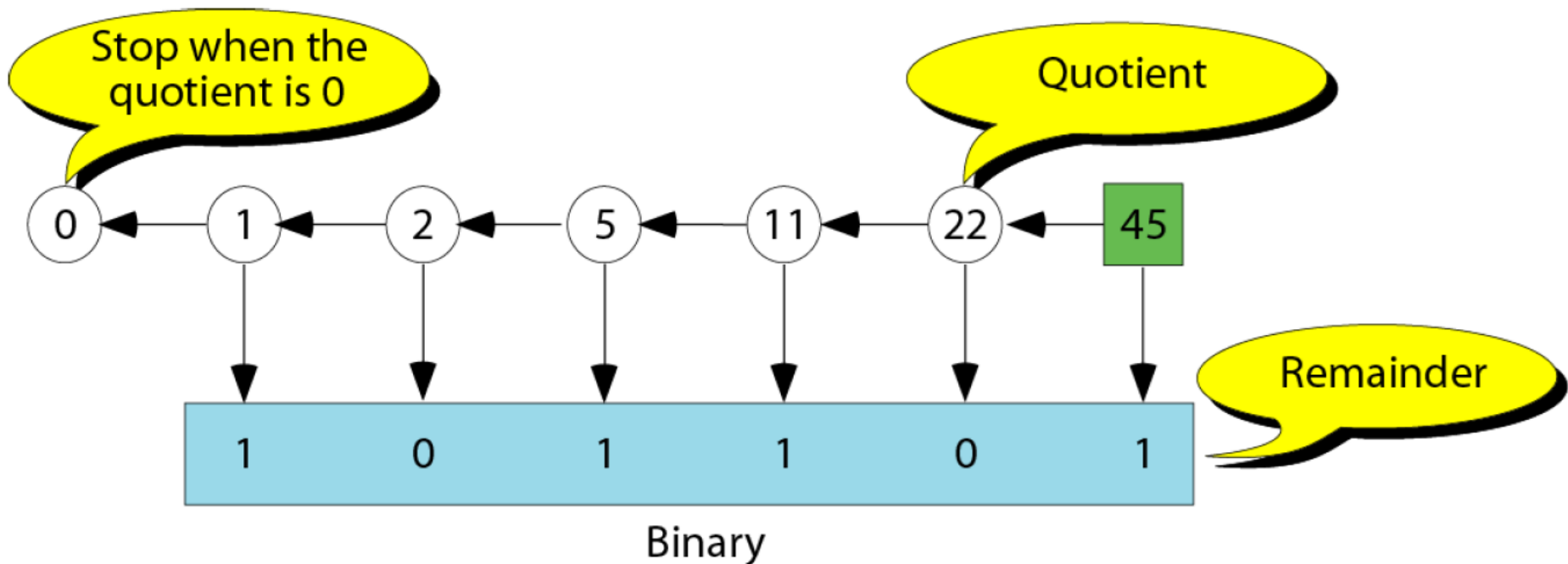
- Repeat division by 2 until the quotient is 0 and keep the remainder. Pass the quotient to the next stage, until it is zero
- E.g. 45 to binary?
- $45/2=22$ r1
- $22/2=11$ r0
- $11/2=5$ r1
- $5/2=2$ r1
- $2/2=1$ r0
- $1/2=0$ r1
- 45 -> 101101

Conversion from Decimal to Binary

- Convert 75 to binary

Decimal to Binary Conversion Example

E.g. start with 45 and at each stage divide by 2 noting the quotient (to pass on) and remainder (bit)



$$45 = 32 \times 1 + 16 \times 0 + 8 \times 1 + 4 \times 1 + 2 \times 0 + 1 \times 1 \text{ or } 101101 \text{ binary}$$

Conversion from Binary to Decimal

- Multiply each number with each corresponding power of 2 and add all the resulting numbers

- E.g. 10010011

$$1 \times 2^7 + 1 \times 2^4 + 1 \times 2^1 + 1 \times 2^0$$

$$1 \times 128 + 1 \times 16 + 1 \times 2 + 1 \times 1 = 147$$

- Check: 147 decimal to binary

$$147/2 = 73 \text{ r}1$$

$$73/2 = 36 \text{ r}1$$

$$36/2 = 18 \text{ r}0$$

$$18/2 = 9 \text{ r}0$$

$$9/2 = 4 \text{ r}1$$

$$4/2 = 2 \text{ r}0$$

$$2/2 = 1 \text{ r}0$$

$$1/2 = 0 \text{ r}1 \text{ so } 147 \text{ binary is } 10010011$$

0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024
11	2048
12	4096
etc.	

Conversion from Binary to Decimal

- Convert 10110101 to decimal

$$(1 \times 2^7) + (0 \times 2^6) + (1 \times 2^5) + (1 \times 2^4) + (0 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (1 \times 2^0)$$

$$(1 \times 128) + (0 \times 64) + (1 \times 32) + (1 \times 16) + (0 \times 8) + (1 \times 4) + (0 \times 2) + (1 \times 1)$$

$$128 + 0 + 32 + 16 + 0 + 4 + 0 + 1 = 181$$

0	1
1	2
2	4
3	8
4	16
5	32
6	64
7	128
8	256
9	512
10	1024
11	2048
12	4096
etc.	

Conversion from Binary to Hexadecimal

- Writing and reading binary numbers is tedious and error prone. Hexadecimal is a perfect number system for representing 4 bits
- So with a large binary number, divide into 4 bits starting from the right (lsb) and write down the corresponding hex representation
- e.g.
- 111110101001110100100101010
- 0111 1101 0100 1110 1001 0010 1010
- 7 D 4 E 9 2 A
- and back again is just as easy:
- E.g. C4DE9 = 1100 0100 1101 1110 1001

Conversion from Binary to Hexadecimal

1011110001011

0001 0111 1000 1011

1 7 8 B

0x178B

Practical Session

- Make sure you know how to use Brookes Moodle Site and the computing environment for the module
- A quiz on-line to practice binary conversion

At home

- Complete the exercises and submit them online
- Read chapter 1 of your book for a history of computers
- Read first 10 chapters of Code Petzold's book to understand how binary has been used in history and to see more examples of not 10-base numbers